

Apply filters to SQL queries

Project description

The security team at my organization needs to investigate some potential security issues that involve login attempts and employee machines. The login attempts and employees tables need to be examined in order to reveal these security issues. SQL and filters will be used to retrieve records from different data sets in these tables to investigate.

Retrieve after hours failed login attempts

The following SQL code shows how I determined the login attempts made after hours.

```
MariaDB [organization]> Select * From log_in_attempts Where login_time > '18:00:00' AND Success = 0;
```

event_id	username	login_date	login_time	country	ip_address
205.12	2 apatel	2022-05-10	20:27:27	CAN	192.168.
66.142	18 pwashing	2022-05-11	19:28:50	US	192.168.
109.50	20 tshah	2022-05-12	18:56:36	MEXICO	192.168.
27.57	28 aestrada	2022-05-09	19:28:12	MEXICO	192.168.
45.93	34 drosas	2022-05-11	21:02:04	US	192.168.
4.157	42 cgriffin	2022-05-09	23:04:05	US	192.168.
58.57	52 cjackson	2022-05-10	22:07:07	CAN	192.168.
100.17	69 wjaffrey	2022-05-11	19:55:15	USA	192.168.
234.49	82 abernard	2022-05-12	23:38:46	MEX	192.168.
132.153	87 apatel	2022-05-08	22:38:31	CANADA	192.168.
	96 ivelasco	2022-05-09	22:36:36	CAN	192.168.

Lines 1 - 2 display the code I entered to filter the failed login attempts that occurred after hours in the login attempts table with SQL. The lines following this display the login attempts table with these filters. I used the Select keyword and * wildcard symbol to select all the columns of the login attempts table to be in the output. I used the Where clause to specify which column I want to filter and the > symbol to specify after hours. Following that is the AND operator which was used to add another condition that must be true Success = 0. The output indicates that there were 19 failed login attempts after 18:00.

Retrieve login attempts on specific dates

The following SQL code demonstrates how I keywords, clauses, and certain operators to retrieve the login attempts made on specific dates.

```
MariaDB [organization]> Select * From log_in_attempts Where login_date = '2022-05-08' OR login_date = '2022-05-09';
```

event_id	username	login_date	login_time	country	ip_address	success
243.140	jrafael	2022-05-09	04:56:27	CAN	192.168.140	1
151.162	dkot	2022-05-09	06:47:41	USA	192.168.162	1
178.71	dkot	2022-05-08	02:00:39	USA	192.168.71	0
119.173	bisles	2022-05-08	01:30:17	US	192.168.173	0
100.158	dkot	2022-05-08	09:11:34	USA	192.168.158	1
183.51	lyamamot	2022-05-09	17:17:26	USA	192.168.51	0
171.192	arusso	2022-05-09	06:49:39	MEXICO	192.168.192	1
33.137	sbaelish	2022-05-09	07:04:02	US	192.168.137	1
123.105	apatel	2022-05-08	17:27:00	CANADA	192.168.105	1
27.57	aestrada	2022-05-09	19:28:12	MEXICO	192.168.57	0
	yappiah	2022-05-09	03:22:22	MEX	192.168.57	1

Lines 1 - 2 display the code I entered to filter the login attempts made on specific days with SQL and show it on the login attempts table. The lines following this display the login attempts table with these filters. I used the Select keyword and * wildcard symbol to select all the columns of the login attempts table to be in the output. I used the Where clause to specify which column I want to filter "login_date" and the OR operator to display 2 different dates which are 2022-05-08 and 2022-05-09. The output indicates at the end that there were 75 login attempts made on these 2 days in total.

Retrieve login attempts outside of Mexico

After investigating the organization's data on login attempts, I believe there is an issue with the login attempts that occurred outside of Mexico. These login attempts should be investigated.

The following code demonstrates how I created a SQL query to filter for login attempts that occurred outside of Mexico.

```
MariaDB [organization]> Select * From log_in_attempts Where Not coun
try Like 'MEX%';
+-----+-----+-----+-----+-----+-----+
| event_id | username | login_date | login_time | country | ip_address |
| success |
+-----+-----+-----+-----+-----+-----+
| 1 | jrafael | 2022-05-09 | 04:56:27 | CAN | 192.168.243.140 |
| 1 |
| 2 | apatel | 2022-05-10 | 20:27:27 | CAN | 192.168.205.12 |
| 0 |
| 3 | dkot | 2022-05-09 | 06:47:41 | USA | 192.168.151.162 |
| 1 |
| 4 | dkot | 2022-05-08 | 02:00:39 | USA | 192.168.178.71 |
| 0 |
| 5 | jrafael | 2022-05-11 | 03:05:59 | CANADA | 192.168.86.232 |
| 0 |
| 7 | eraab | 2022-05-11 | 01:45:14 | CAN | 192.168.
```

The first part of the screenshot is my query, and the second part is a portion of the output. This query returns all login attempts that occurred in countries other than Mexico. First, I started by selecting all data from the `log_in_attempts` table. Then, I used a `WHERE` clause with `NOT` to filter for countries other than Mexico. I used `LIKE` with `MEX%` as the pattern to match because the dataset represents Mexico as `MEX` and `MEXICO`. The percentage sign (%) represents any number of unspecified characters when used with `LIKE`.

Retrieve employees in Marketing

My team wants to update the computers for certain employees in the Marketing department. To do this, I have to get information on which employee machines to update.

The following code demonstrates how I created a SQL query to filter for employee machines from employees in the Marketing department in the East building.

```
MariaDB [organization]> Select * From employees Where department = '
Marketing' And office Like 'East%';
+-----+-----+-----+-----+-----+
| employee_id | device_id | username | department | office |
+-----+-----+-----+-----+-----+
| 1000 | a320b137c219 | elarson | Marketing | East-170 |
| 1052 | a192b174c940 | jdarosa | Marketing | East-195 |
| 1075 | x573y883z772 | fbautist | Marketing | East-267 |
| 1088 | k865l965m233 | rgosh | Marketing | East-157 |
| 1103 | NULL | randers | Marketing | East-460 |
| 1156 | a184b775c707 | dellery | Marketing | East-417 |
| 1163 | h679i515j339 | cwilliam | Marketing | East-216 |
+-----+-----+-----+-----+-----+
7 rows in set (0.001 sec)
```

The first part of the screenshot is my query, and the second part is a portion of the output. This query returns all employees in the Marketing department in the East building. First, I started by selecting all data from the `employees` table. Then, I used a `WHERE` clause with `AND` to filter for employees who work in the Marketing department and in the East building. I used `LIKE` with `East%` as the pattern to match because the data in the `office` column represents the East building with the specific office number. The first condition is the `department = 'Marketing'` portion, which filters for employees in the Marketing department. The second condition is the `office LIKE 'East%'` portion, which filters for employees in the East building.

Retrieve employees in Finance or Sales

The machines for employees in the Finance and Sales departments also need to be updated. Since a different security update is needed, I have to get information on employees only from these two departments.

The following code demonstrates how I created a SQL query to filter for employee machines from employees in the Finance or Sales departments.

```
MariaDB [organization]> Select * From employees Where department = 'Finance' OR department = 'Sales';
```

employee_id	device_id	username	department	office
1003	d394e816f943	sgilmore	Finance	South-153
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodriqu	Sales	South-134
1010	k242l212m542	jlansky	Finance	South-109
1011	l748m120n401	drosas	Sales	South-292
1015	l611n260w045	lrodriqu	Finance	North-071

The first part of the screenshot is my query, and the second part is a portion of the output. This query returns all employees in the Finance and Sales departments. First, I started by selecting all data from the `employees` table. Then, I used a `WHERE` clause with `OR` to filter for employees who are in the Finance and Sales departments. I used the `OR` operator instead of `AND` because I want all employees who are in either department. The first condition is `department = 'Finance'`, which filters for employees from the Finance department. The second condition is `department = 'Sales'`, which filters for employees from the Sales department.

Retrieve all employees not in IT

My team needs to make one more security update on employees who are not in the Information Technology department. To make the update, I first have to get information on these employees.

The following demonstrates how I created a SQL query to filter for employee machines from employees not in the Information Technology department.

```
MariaDB [organization]> Select * From employees Where NOT department = 'Information Technology';
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1001	b239c825d303	bmoreno	Marketing	Central-276
1002	c116d593e558	tshah	Human Resources	North-434
1003	d394e816f943	sgilmore	Finance	South-15

The first part of the screenshot is my query, and the second part is a portion of the output. The query returns all employees not in the Information Technology department. First, I started by selecting all data from the `employees` table. Then, I used a `WHERE` clause with `NOT` to filter for employees not in this department.

Summary

I applied filters to SQL queries to get specific information on login attempts and employee machines. I used two different tables, `log_in_attempts` and `employees`. I used the `AND`, `OR`, and `NOT` operators to filter for the specific information needed for each task. I also used `LIKE` and the percentage sign (%) wildcard to filter for patterns.